GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 17



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Recap of Previous Lecture



Topic Mensuration 2D & 3D

Topics to be Covered

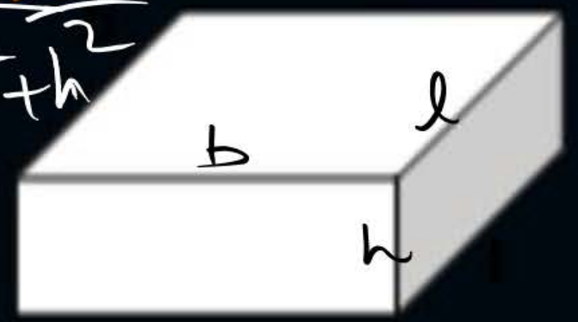


Topic-1 Interest & Sequence

MENSURATION: 3D

Longest diagonal

Cuboid



$$\text{Lateral Surface} = \sqrt{l^2+b^2+h^2}$$

$$\text{Area} = 2h(l+b)$$

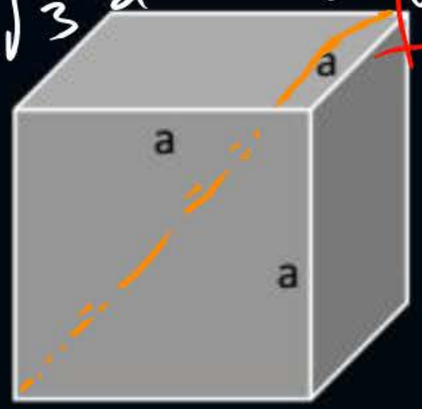
$$\text{Total Surface Area} = 2h(l+b)+2lb = 2(lb+bh+hl)$$

$$\text{Volume} = lbh$$

Cube:

Longest diagonal $= \sqrt{a^2 + a^2 + a^2}$

$= \sqrt{3a^2} = \sqrt{3}\,a$



Lateral Surface Area $= 4a^2$

Total Surface Area $= 4a^2 + 2a^2$

$= 6a^2$

Volume $= a^3$

Cylinder:

Volume $= \pi r^2 h$

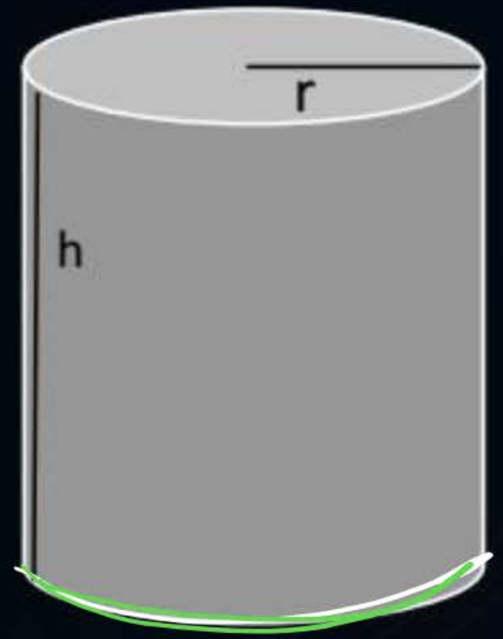
Curved Surface Area

$= 2\pi r h$

Total Surface Area

$= 2\pi r h + 2\pi r^2$

$= 2\pi r (h + r)$



Cone:

$$\text{Volume} = \frac{1}{3}\pi r^2 h$$

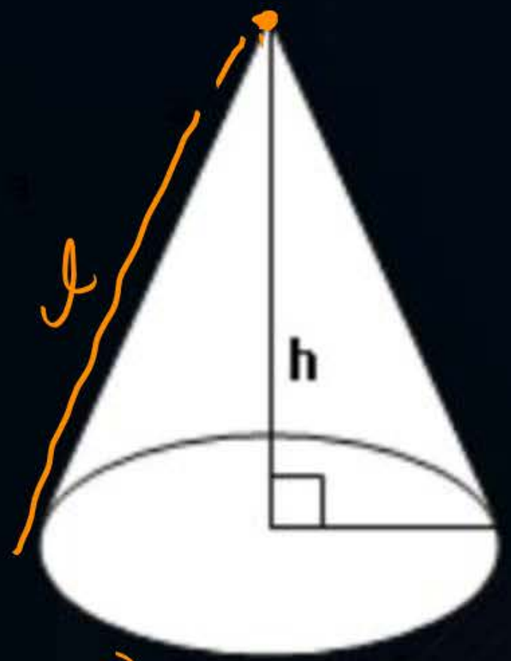
Curved Surface

$$\text{Area} = \pi r l$$

Total Surface Area

$$= \pi r l + \pi r^2$$

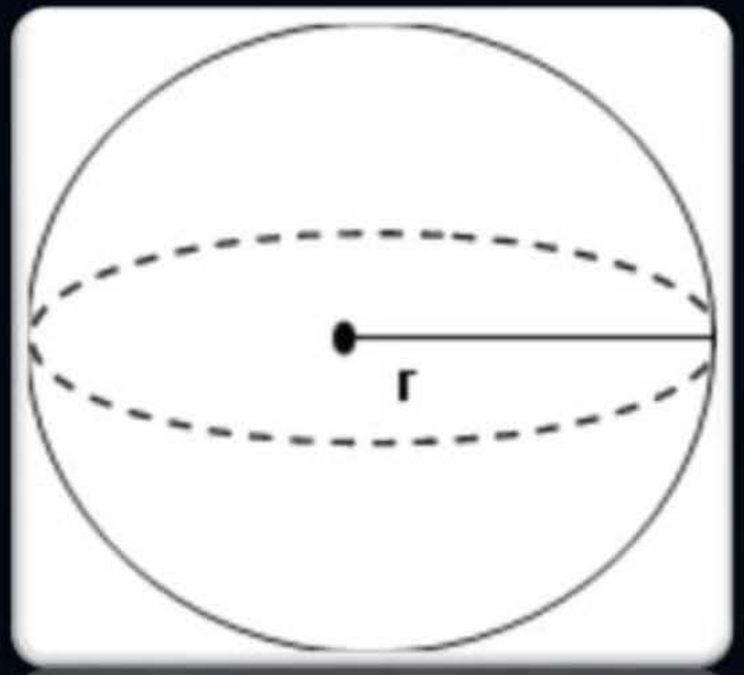
$$= \pi r (l + r)$$



Sphere:

Surface Area $= 4\pi r^2$

Volume $= \frac{4}{3}\pi r^3$



Hemi-sphere:

$$\text{Volume} = \frac{1}{2} \times \frac{4}{3}\pi r^3$$

$$= \boxed{\frac{2}{3}\pi r^3}$$

Curved Surface Area

$$= 2\pi r^2$$

Total Surface Area

$$= 2\pi r^2 + \pi r^2$$

$$= \underline{3\pi r^2}$$



[MCQ]

#Q. What would be the area of sector of a circle whose radius is 12 cm and the length of the arc is 20 cm?

Assignment

$120\ cm^2$

Area of Sector $= \frac{\theta}{360} \times \pi r^2$

l of arc $= \frac{\theta}{360} \times 2\pi r$

$20 = \frac{\theta}{360} \times 2\pi r$

$12 \times 10 = \frac{\theta}{360} \times \pi r \times r$

A 60 sq. cm

B 240 sq. cm

C 120 sq. cm

D 64 sq. cm

[MCQ]

#Q. The cross-section of a canal is a trapezium in shape. If the canal is 7 m wide at the top and 9 m at the bottom and the area of cross-section is 128 sq m, find the height of the canal.

Assignment

A 32 m

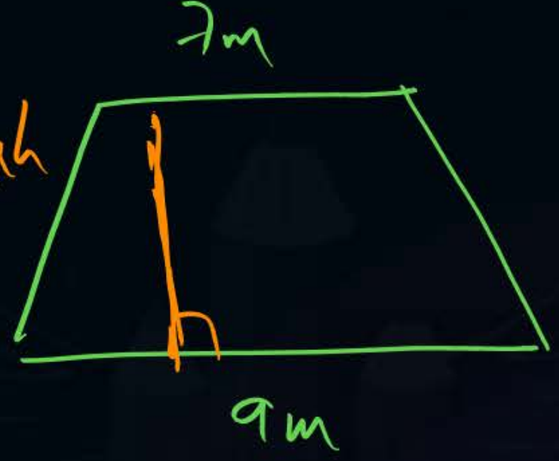
B 8 m

C 16 m

D 4 m

Area of Trapezium

$= \frac{1}{2}$(sum of 11 sides) $\times h$



$128 = \frac{1}{2} \times 16 \times h$

$\frac{128}{8} = h = 16\ m$

[MCQ]

#Q. A cube of 6 cm side melted and smaller cubes of 2 cm side are manufactured. Find the number of smaller cubes so formed.

Assignment

$$\text{Volume} = a^3$$

A 12

B 27

C 24

D 8

$$\text{Big Volume} = 6^3 = 216$$

$$\text{Small cube Volume} = 2^3 = 8$$

$$\frac{216}{8} = 27$$

[MCQ]

#Q. The length, breadth & height of a cuboid are in the ratio of 4 : 3 : 2 and its volume is 3000 m^3. Find its total surface area.



$l = 4x$; $b = 3x$; $h = 2x$

Volume cuboid $= lbh$

$3000 = 4x \times 3x \times 2x$

$3000 = 24x^3$

$x^3 = \frac{3000}{24} = 125$

$x = 5$

Assignment

Total Surface Area

$= 2(lb + bh + hl)$

$= 2 \times (300 + 150 + 200)$

$= 2 \times 650$

$= 1300\ m^2$

A. 1300 m^2

B. 1500 m^2

C. 1333 m^2

D. 2700 m^2

[MCQ]

l b h

#Q. The dimensions of a room are 20 m x 15 m x 10 m. What is the cost of painting its four walls at the rate of ₹ 15 per 100 sq m?

Assignment

Lateral Surface Area

$= 2h(l+b)$

$= 2 \times 10 (35)$

$= 20 \times 35 = 700 \text{ m}^2$

Cost $= 7 \times 15$

$= ₹ 105$

A ₹ 70

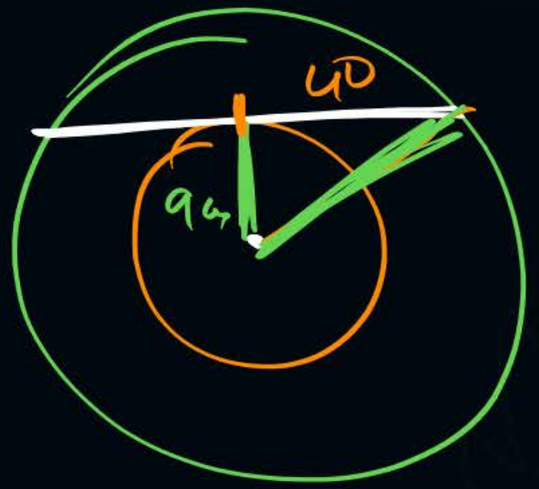
B ₹ 105

C ₹ 225

D None of these

[MCQ]

#Q. In two concentric circles, a chord length 80 cm of larger circle becomes a tangent to the smaller circle whose radius is 9 cm. The radius of the larger circle will be?





Assignment

$h^2 = 81 + 1600$

$h = \sqrt{1681} = 41$ cm

A 13 cm

B 41 cm

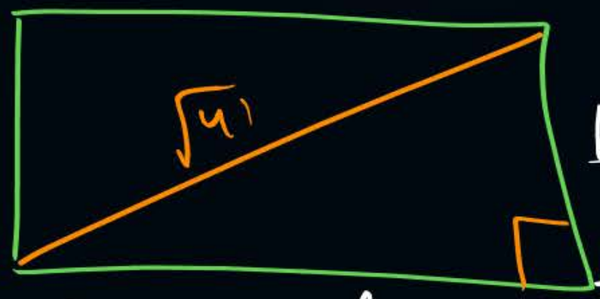
C 52 cm

D 75 cm

[MCQ]

#Q. The diagonal of a rectangle is $\sqrt{41}$ cm and its area is 20 sq cm. The perimeter of the rectangle is?

Assignment



$l \times b = 20$

A 9 cm

B 18 cm

C 20 cm

D 41 cm

$(\sqrt{41})^2 = l^2 + b^2$

$41 = l^2 + b^2$

$l + b = 9$

$P = 2(l+b) = 2 \times 9$

$= 18$ cm

$(l+b)^2 = l^2 + b^2 + 2lb$

$(l+b)^2 = 41 + (2 \times 20) = 81$

INTEREST

What

Why ?

P + I = Amount

100 + 10 = 110

Simple 120

Compound 121

Simple Interest

$$A = P + I$$

$$I = \frac{TRP}{100}$$

$P+$ ↗ ↖ $+P$

$$A = P\left(1 + \frac{TR}{100}\right)$$

Compound Interest

n ?

T = 3 year

n = 3 Annually

n = 6 Half Year

n = 12 Quat

$I = P\left(1+\frac{r}{100}\right)^{n} - P$

$A = P\left(1+\frac{r}{100}\right)^{n}$

$I = P\left[\left(1+\frac{r}{100}\right)^{n} - 1\right]$

Note:

1. How to understand S.I or C.I?
2. Time can be broken (S.I)
3. Maximum (C.I)

Further formula:

2 yrs

Difference

$$P\left(\frac{r}{100}\right)^2$$

C.I − S.I

$$P\left(1+\frac{r}{100}\right)^2 - P\left(1+\frac{2r}{100}\right)$$

$$P\left[1+\left(\frac{r}{100}\right)^2+\frac{2r}{100}-1-\frac{2r}{100}\right]$$

[MCQ]

#Q. An investment triples itself in 10years. In how many years it would be nine times itself?

Amount = 3P → 10yrs

I = 2P (×4) → 8P (×4) → 40yrs

I

A = 9P

$A = P\left(1+\frac{TR}{100}\right)$

$3P = P\left(1+\frac{10R}{100}\right)$

$2 = \frac{10R}{100}$

R = 20%

$9P = P\left(1+\frac{T\times 20}{100\times 5}\right)$

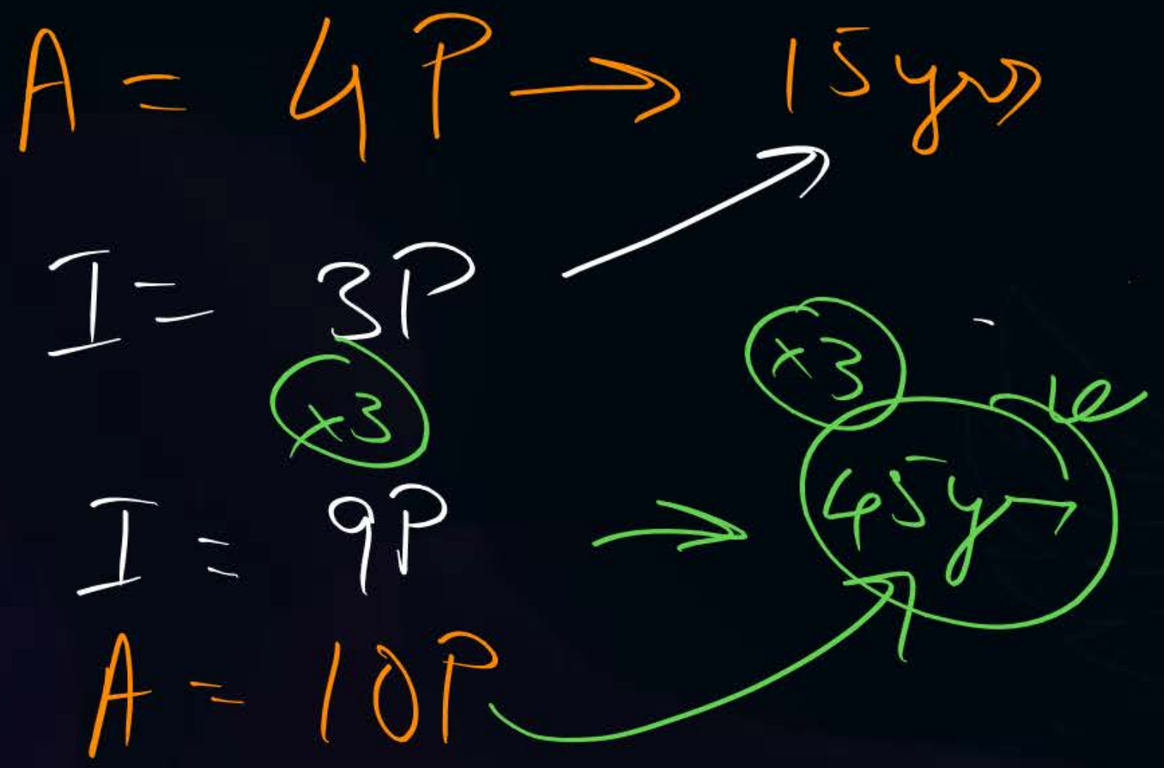
$8 = \frac{T}{5}$

T = 40yrs

[MCQ]

4 times

#Q. An investment quadruple itself in 15years. Then in how many years it would be ten times itself?



[MCQ]

#Q. What is the compound interest on ₹2500 for two years at the rate of interest 4% per annum?

Assignment

A ₹ 180

B ₹ 210

C ₹ 204

D ₹ 220

[MCQ]

#Q. A sum of money was invested at simple interest at a certain rate for 3 years. Had it been invested at a 4% higher rate, it would have fetched ₹480 more. Find the principal.

Assignment

A ₹ 3000

B ₹ 4000

C ₹ 5000

D ₹ 6000

[MCQ]

#Q. In what time a sum of money will double itself at a rate of S.I. of 6% per annum?

Assignment

A. 6 ⅓ years

B. 16 ⅔ years

C. 33 ⅓ years

D. 21 years

[MCQ]

#Q. In how many years will a sum of ₹800 at 10% per annum compounded semi-annually become ₹926.10?

Assignment

A. 1 ⅓ years

B. 1 ½ years

C. 2 ⅓ years

D. 2 ½ years

[MCQ]

#Q. The difference between simple interest and compound interest on a sum for 2 years at 8% when the interest is compounded annually is ₹16. If the interest were compounded half yearly the difference in two interests would be nearly

Assignment

A ₹ 21.25

B ₹ 24.64

C ₹ 27.85

D ₹ 29.94

[MCQ]

#Q. Raju invested Rs.5488 for 3 years at C.I. and received an amount of Rs.6750 on maturity. What is rate percent?

Assignment

A. $7\frac{1}{7}\%$

B. $9\frac{2}{3}\%$

C. $3\frac{1}{3}\%$

D. $6\frac{1}{3}\%$



2 mins Summary

Topic Interest & Sequence

Simple

Logical

Compound

?

THANK - YOU